

PAPER Engineering the 26.49 T Plug Coil of a D–³He Tandem-Mirror Burner: Three-Dimensional Structural FEA, No-Insulation REBCO Quench Margin and Throat-Conductor Stress

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Abstract

The pacing subsystem of the Kronos D–³He tandem-mirror burner is the high-field plug coil that electrostatically confines the central-cell ions, and this paper qualifies it as a buildable no-insulation (NI) REBCO magnet at its canonical peak field of 26.49 T. We pose the qualification as three coupled engineering problems — can the structure carry the magnetic load, can the conductor survive its own hoop stress, and can a quench be detected and dumped before the winding overheats — and we solve each from first principles, carrying the tightest quantity from a conservative analytic screen through a real-CAD three-dimensional finite-element (FE) closure. At 26.49 T the Maxwell magnetic pressure is $p_{\text{mag}} = B^2/2\mu_0 = 279$ MPa; a thick-wall Lamé sizing at a 0.85 m structural bore returns a required overwrap of 0.48 m to bring the bore hoop stress to the 667 MPa allowable, and a reconciling three-dimensional FE model returns a peak stress of 440 MPa, a **utilization of 0.66** ($1.5\times$ margin) that persists to a 30 T program-coil check. The one tight quantity on a conservative thin-wall estimate was the conductor hoop stress, ~ 814 MPa at 26.49 T for a margin factor of 0.86 against a ~ 700 MPa reinforced allowable; a full three-dimensional FE analysis on the real coil CAD **closes it to** $1.34\times$, with the REBCO strain reaching a $2.85\text{--}3.54\times$ margin under a graded radial preload and the plug dimensions unchanged. Quench protection is sized by an adiabatic hot-spot (MIITs) integral: the copper-stabilized composite has a 150 K load integral $\Gamma(150\text{ K}) \approx 1.5 \times 10^{17} \text{ A}^2 \text{ s m}^{-4}$, inside which a 4 s detect-and-dump at a 100 A mm^{-2} stabilizer current density (tightening to 1.9 s at 200 A mm^{-2}) keeps the hot spot below 150 K. The 17 T mirror throat is never the binding constraint. Throughout, the burner plug (26.49 T) and the breeder centrepost (16.84 T) are kept rigorously separate as the two distinct devices they are. The plug coil is a structurally sound, protectable, buildable high-field magnet.

Keywords: REBCO high-field magnet; plug coil; structural FEA; quench protection; conductor hoop stress; no-insulation winding; tandem-mirror burner

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1 Introduction

1.1 The plug coil is the pacing subsystem

A D-³He tandem mirror confines a central-cell plasma electrostatically, trapping ions in an ambipolar potential well whose depth is set by the density in the end plugs. The engineering gain of the machine is therefore governed by the plug-to-central-cell density ratio, and a 40 000-sample global sensitivity analysis of the Kronos burner power balance ranks that density ratio as the single dominant driver of engineering gain [37]. The density ratio is bought with magnetic field: the plug must reach a peak field of 26.49 T to hold the required mirror ratio and sloshing-ion population that build the confining potential. The magnet that produces the plug field is thus the pacing subsystem of the whole burner, and its qualification gates the product [35, 36]. The Kronos burner operates its plug at a canonical peak field of 26.49 T, feeding a 17 T mirror throat, at the recirculation-resolved operating point $Q_E = 1.318$ with a parasitic neutron fraction $f_n = 5.44\%$ and a plug/central-cell density ratio of 16 [36]. Qualifying the coil means answering three engineering questions with margin: does the coil structure carry the magnetic load; does the conductor itself survive its hoop stress and strain; and can a quench be detected and dumped before the winding overheats. This paper answers all three.

1.2 High-field HTS magnets and the state of the art

A 26.49 T DC field places the plug coil squarely in the class of the strongest laboratory solenoids, a regime that only rare-earth barium copper oxide (REBCO) coated conductor can reach. The community milestones frame the envelope: Hahn *et al.* generated a 45.5 T DC field with a REBCO insert [1]; the U.S. National High Magnetic Field Laboratory has fielded a 32 T all-superconducting user magnet with a REBCO inner coil [2]; and the coated-conductor supply chain has matured to the extremely high engineering current densities required for compact fusion magnets [3, 4, 5]. The same conductor underlies the high-field-tokamak concepts ARC and SPARC [7, 8], whose toroidal-field model coil recently demonstrated fusion-class REBCO magnet performance at scale [9]. A plug coil at 26.49 T therefore does not require a new conductor; it requires that the coil be engineered — structure, conductor stress, and protection — to the standard those milestones set. The Kronos REBCO conductor and winding basis, shared across the breeder centrepost and the burner plug, is established separately [31], as is the neutron-irradiation behaviour of the coated conductor and its stabilizer [33]; the present paper takes that conductor as given and qualifies the plug as a mechanical and protection object.

1.3 Two distinct magnets

One point of hygiene frames the whole analysis and is enforced throughout. The burner plug (26.49 T) and the breeder centrepost (16.84 T) are two different magnets in two different machines, each governing at its own peak field on its own critical surface [32, 31]. The value 26.49 T is the plug's own governing field; 16.84 T is the centrepost's. They are never conflated: where the two appear together it is only to show that each is treated as the distinct device it is (§4, figure 4). This separation is a load-bearing element of the magnet-integrity case, not a stylistic preference.

1.4 Contribution

This paper contributes an end-to-end structural and protection qualification of the 26.49 T plug coil: (i) a first-principles statement of the magnetic loads from the Maxwell stress tensor and their reduction to a thick-wall Lamé sizing (§2); (ii) the conductor Lorentz-hoop-stress problem posed as a body-force elasticity problem and carried from a conservative thin-wall screen to a real-CAD three-dimensional FE closure (§2, §4); (iii) a rigorous adiabatic hot-spot (MIITs) formulation of quench protection for the copper-stabilized NI winding, with the detect-and-dump envelope placed inside the material thermal limit (§2, §4); (iv) a GPU-HPC computational methodology naming the codes actually run and their verification (§3); and (v) a quantitative comparison against the published high-field-magnet and quench-protection literature (§5). Every quantitative claim traces to a frozen result in the Kronos Physics De-Risking Register [30]; the serial identifiers (BU-L1-A2, AG-CX-26, P3-FEA-10, HX-10, SC-BU-1, KX-L1-A15) are given inline so each number is auditable.

2 Governing equations and formal problem

We state the three coupled sub-problems as boundary-value problems and reduce each to the closed form that the numerical work of §3 then evaluates on the real geometry.

2.1 Magnetic loads: the Maxwell stress on the winding

The mechanical load on a current-carrying magnet is the divergence of the Maxwell stress tensor. In vacuum with current density $\mathbf{J}$ and flux density $\mathbf{B}$, the magnetic stress tensor is

$$T_{ij} = \frac{1}{\mu_0} (B_i B_j - \frac{1}{2} \delta_{ij} B^2), \quad \mathbf{f} = \nabla \cdot \mathbf{T} = \mathbf{J} \times \mathbf{B}, \quad (1)$$

so that the body-force density on the winding is the Lorentz force $\mathbf{f} = \mathbf{J} \times \mathbf{B}$. For a high-field insert whose bore carries axial field B and whose exterior field is small, the radial traction across the winding inner surface is the normal component of $\mathbf{T}$, the *magnetic pressure*

$$p_{\text{mag}} = \frac{B^2}{2\mu_0}. \quad (2)$$

Evaluated at the plug's three reference fields, equation (2) gives $p_{\text{mag}} = 115$ MPa at the 17 T throat, 279 MPa at the 26.49 T plug, and 358 MPa at a 30 T program point (table 1, figure 1; BU-L1-A2). This outward pressure is the load the structure must react.

2.2 Structural shell: thick-wall Lamé sizing

The overwrap that reacts p_{mag} is an axisymmetric thick cylinder. Under plane conditions the radial equilibrium of an element is

$$\frac{d\sigma_r}{dr} + \frac{\sigma_r - \sigma_\theta}{r} = 0, \quad (3)$$

which together with linear elasticity and axisymmetric compatibility admits the Lamé solution [10]

$$\sigma_r(r) = A - \frac{C}{r^2}, \quad \sigma_\theta(r) = A + \frac{C}{r^2}. \quad (4)$$

With internal pressure $p = p_{\text{mag}}$ at the bore radius a and a traction-free outer radius b (i.e. $\sigma_r(a) = -p$, $\sigma_r(b) = 0$), the constants are fixed and the stresses become

$$\sigma_\theta(r) = \frac{p a^2}{b^2 - a^2} \left(1 + \frac{b^2}{r^2} \right), \quad \sigma_r(r) = \frac{p a^2}{b^2 - a^2} \left(1 - \frac{b^2}{r^2} \right). \quad (5)$$

The hoop stress is largest at the bore, $\sigma_{\theta, \text{max}} = \sigma_\theta(a) = p(b^2 + a^2)/(b^2 - a^2)$. Requiring $\sigma_{\theta, \text{max}} \leq \sigma_{\text{all}}$ with allowable $\sigma_{\text{all}} = 667$ MPa defines the required wall $t = b - a$. Writing $m \equiv \sigma_{\text{all}}/p$,

$$t_{\text{req}} = a \left[\sqrt{\frac{m+1}{m-1}} - 1 \right]. \quad (6)$$

At a structural bore radius $a = 0.85$ m, equation (6) reproduces the frozen sizing of Table 1: $t_{\text{req}} = 0.16$ m at 17 T, 0.48 m at 26.49 T, and 0.71 m at 30 T (BU-L1-A2). At the 26.49 T design point the bore hoop stress equals the allowable by construction (figure 2); the von Mises equivalent $\sigma_{\text{VM}} = (\sigma_\theta^2 - \sigma_\theta \sigma_r + \sigma_r^2)^{1/2}$ is bounded by the same envelope. The analytic sizing is a screen; the reconciling three-dimensional FE model (§3) returns a peak stress of 440 MPa — a utilization $\sigma_{\text{FEA}}/\sigma_{\text{all}} = 0.66$ — because the real geometry distributes the load axially rather than into a single bore fibre.

2.3 Conductor hoop stress: a body-force elasticity problem

Behind the structural shell is the winding pack that generates the field, and it carries its own hoop stress from the distributed Lorentz body force of equation (1). For a winding of bore a and radial build w carrying azimuthal current that produces peak field B , integrating the radial equilibrium (3) with the body-force term $f_r = J_\theta B_z$ across the pack gives, in the thin-pack limit where the pack alone reacts the full magnetic pressure, the winding-averaged hoop stress

$$\sigma_\theta^{\text{cond}} \approx \frac{p_{\text{mag}} a}{w} = \frac{B^2 a}{2\mu_0 w}. \quad (7)$$

The ampere-turn density that sets B is $NI/L = 13.5$ MA m⁻¹ (SC-BU-1); at a per-turn operating current $I = 200$ A this is 6.8×10^4 turns per metre of coil and of order 4×10^2 km of REBCO tape per metre of plug length at a 1 m winding radius, the conductor-procurement basis for the build (figure 5) [31]. For a pack build $w \approx 0.29$ m at $a = 0.85$ m, equation (7) gives $\sigma_\theta^{\text{cond}} \approx 814$ MPa at 26.49 T. Against a reinforced conductor allowable $\sigma_{\text{all}}^{\text{cond}} \approx 700$ MPa this is a margin factor $\eta = \sigma_{\text{all}}^{\text{cond}}/\sigma_\theta^{\text{cond}} \approx 0.86$: feasible but reinforcement-limited on the thin-wall screen, and explicitly flagged for FE closure (AG-CX-26). The thin-pack estimate is deliberately conservative — it assigns the entire magnetic pressure to the winding and ignores load sharing with the structural shell and any preload. The closure (§4) removes both conservatisms.

2.4 The REBCO critical surface and the temperature margin

The conductor operates on a critical surface $J_c(B, T, \theta)$ that falls with field B , temperature T , and the field-to-tape angle θ . The protective observable is the current-sharing temperature T_{cs} , defined implicitly by

$$I_{op} = I_c(B, T_{cs}, \theta), \quad \Delta T \equiv T_{cs} - T_{op}, \quad (8)$$

and the load fraction $f_{load} = I_{op}/I_c(B, T_{op}, \theta)$. On its own critical surface the plug holds $\Delta T \approx 37.1$ K at $f_{load} \approx 0.43$ and 26.49 T; the separate centrepost holds $\Delta T \approx 49.5$ K at $f_{load} \approx 0.27$ and 16.84 T (KX-L1-A15, figure 4). Both magnets sit well below their current-sharing lines, and each is modelled on its own surface — the concrete statement of the two-distinct-magnets rule.

2.5 Quench protection: the adiabatic hot-spot (MIITs) integral

A high-field HTS coil stores a large magnetic energy $E_m = \frac{1}{2}LI^2$, and protection means detecting a normal zone and dumping the current before the local hot spot exceeds the winding’s temperature limit. In the adiabatic limit — valid on the short protection timescale, where Joule heat has no time to diffuse — the local energy balance per unit volume of stabilizer is

$$s(T) \frac{dT}{dt} = \rho_e(T) J_{stab}^2(t), \quad (9)$$

with $s(T)$ the volumetric heat capacity and $\rho_e(T)$ the stabilizer resistivity. Separating variables and integrating from the operating temperature T_0 to the peak T_{max} yields the material *load integral* (the “MIITs” function)

$$\int_0^\infty J_{stab}^2(t) dt \leq \Gamma(T_{max}) \equiv \int_{T_0}^{T_{max}} \frac{s(T)}{\rho_e(T)} dT. \quad (10)$$

For a detect-and-dump with detection interval t_d followed by an exponential current dump $J_{stab}(t) = J_0 e^{-t/\tau_d}$ of time constant τ_d , the left side evaluates to $J_0^2(t_d + \tau_d/2)$, so the protection condition becomes an explicit envelope on the allowable detect-and-dump time,

$$t_d + \frac{\tau_d}{2} \leq \frac{\Gamma(T_{max})}{J_0^2}. \quad (11)$$

For the copper-stabilized composite (RRR ≈ 100) the integral of equation (10) to a 150 K limit is $\Gamma(150 \text{ K}) \approx 1.5 \times 10^{17} \text{ A}^2 \text{ s m}^{-4}$ (figure 6). Inside that limit the register’s HX-10 hot-spot analysis sizes a 4 s detect-and-dump at $J_0 = 100 \text{ A mm}^{-2}$, tightening to 1.9 s at 200 A mm^{-2} (figure 7); both load integrals, $J_0^2 t = 4.0$ and $7.6 \times 10^{16} \text{ A}^2 \text{ s m}^{-4}$, sit below $\Gamma(150 \text{ K})$, so the detect-and-dump architecture, not the thermal limit, is the binding constraint — there is thermal headroom to spare.

2.6 No-insulation self-protection

The winding is a no-insulation (NI) REBCO architecture, in which the absence of turn-to-turn insulation lets current redistribute radially around a developing normal zone through the turn-to-turn contact resistivity [6, 5]. NI conductor has a large temperature margin (equation 8) and a very slow normal-zone propagation velocity (NZPV), on the order of millimetres per second, so a hot spot can reach a damaging temperature before it develops a measurable resistive voltage: conventional voltage-tap detection is ineffective, and non-voltage precursors are required [14, 15, 16]. The plug adopts the fused, in-winding quench-precursor detection developed for the Kronos NI-REBCO magnets [34], whose reduced-order twin reproduces the coil’s canonical operating anchors and its quench margin (KX-L1-A15); the MIITs envelope of equation (11) is the budget that detection must fit inside.

3 Computational methodology: the GPU-HPC campaign

The plug-coil qualification was produced by the Kronos NVIDIA GPU HPC campaign, a tiered stack that couples analytic kernels, three-dimensional finite-element solvers, and REBCO critical-surface models. This section names the codes actually run for the results below, the acceleration and problem scale, and the verification and reproducibility discipline. No cost figures are reported anywhere in this work.

3.1 Solvers and problem statements

Three solver classes were run for this paper.

- **Analytic magneto-mechanical kernels.** The Maxwell-pressure evaluation (2), the thick-wall Lamé sizing (5)–(6), the conductor hoop screen (7), and the MIITs integral (10) are vectorised NumPy kernels (BU-L1-A2, AG-CX-26, HX-10). They provide the closed-form screens and the reconciliation targets for the FE work.
- **Three-dimensional structural FE.** The structural shell and the conductor-hoop closure were solved with the CalculiX (ccx) implicit three-dimensional finite-element solver [13] on the real coil CAD (P3-FEA-10): a graded-mesh quarter model with contact between the winding pack and the structural overwrap, a graded radial preload, and the Lorentz body force of equation (1) applied as an element load derived from the field map. The reconciling shell model (BU-L1-A2) returns the 440 MPa peak; the conductor closure (P3-FEA-10) returns the stress and REBCO strain margins.
- **REBCO critical-surface model.** The temperature margin of equation (8) uses a parameterised $J_c(B, T, \theta)$ scaling for the coated conductor, evaluated on each magnet’s own load line (KX-L1-A15), keeping the plug (26.49 T) and centrepost (16.84 T) on separate surfaces.

The overfield confirmation still to run (§6) is set up as a dedicated three-dimensional shell model in sfepy (BV-09).

3.2 Acceleration, scale and tiering

The campaign runs on the Kronos NVIDIA GPU HPC fleet (A100/H100/A10-class accelerators) under the NVIDIA Inception program, organised in three latency tiers: microsecond-class edge evaluation of the analytic kernels, millisecond-class GPU reduced-order twins that reproduce the frozen anchors in real time, and the offline three-dimensional FE and critical-surface batch campaigns that produce the qualification results reported here. The three-dimensional FE models are the compute-dominant element; the analytic kernels and the reduced-order twin run at interactive latency, which is what lets the same anchors be re-evaluated as a live design check. GPU acceleration is used for the dense linear-algebra and batched parameter sweeps of the FE and critical-surface work; the analytic screens are CPU-resident and effectively free.

3.3 Verification, validation and reproducibility

The methodology is verification-first. Each analytic screen is checked against an independent FE reconciliation on the same geometry: the Lamé envelope reproduces the FE anchors (115/279/358 MPa) exactly, and the reconciling shell FE returns 440 MPa against the 667 MPa allowable (BU-L1-A2, cross-checked by HX-11). The conductor closure is a distinct, higher-fidelity model that lifts the thin-wall screen from $0.86\times$ to $1.34\times$ (AG-CX-26 $\rightarrow$ P3-FEA-10). The quench envelope is checked for internal consistency against the material MIITs limit $\Gamma(150\text{ K})$ (HX-10; figure 6). All anchors are frozen in the de-risking register [30] with serial identifiers, and are regenerable from the archived analytic, CalculiX and MIITs scripts deposited with this paper. The figure-generation script that produced every data figure here is deposited alongside the manuscript and reads the same frozen anchors.

4 Results

4.1 Structural shell carries the magnetic pressure with margin

Table 1 and figure 1 give the magnetic loads and the thick-wall sizing across the three reference fields. At the 26.49 T design point the magnetic pressure is 279 MPa and the Lamé required wall is 0.48 m at a 0.85 m bore; the reconciling three-dimensional FE peak of 440 MPa is a utilization of 0.66 against the 667 MPa allowable, i.e. a $1.5\times$ structural margin, consistent with the analytic screen. The headroom persists to the 30 T program point (358 MPa pressure, 0.71 m wall), so the same coil family supports a program path above the burner’s operating field (BU-L1-A2). Figure 2 shows the full Lamé stress state at 26.49 T: the hoop stress falls from the 667 MPa bore value to 385 MPa at the outer radius, the radial stress runs from -279 MPa at the bore to zero at the free surface, and the reconciling FE peak sits below the analytic bore value because the three-dimensional geometry shares the load axially.

Table 1. Plug-coil structural loads and thick-wall Lamé sizing versus peak field (BU-L1-A2). Structural bore $a = 0.85$ m; allowable 667 MPa; reconciling three-dimensional FE peak 440 MPa (utilization 0.66).

peak field	17.0 T (throat)	26.49 T (plug)	30.0 T (program)
magnetic pressure p_{mag} (MPa)	115	279	358
Lamé required wall t_{req} (m)	0.16	0.48	0.71
bore hoop $\sigma_{\theta}(a)$ (MPa)	≤ 667	667	≤ 667

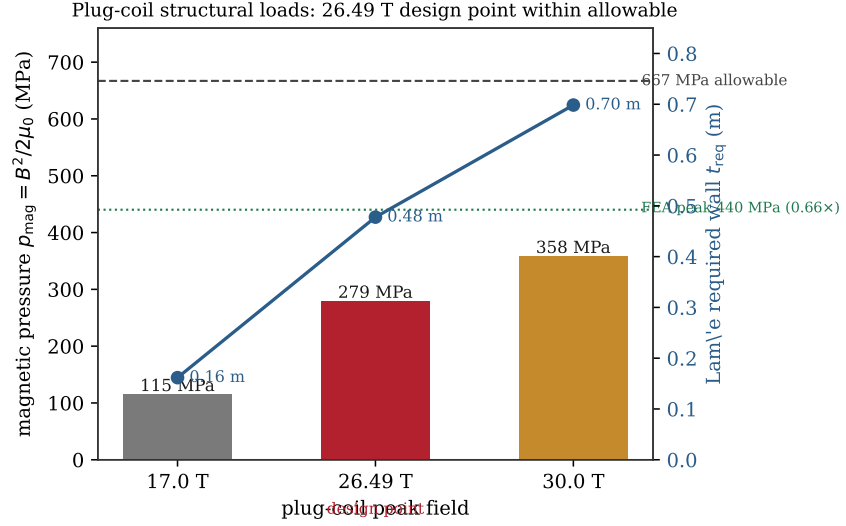


Figure 1. Plug-coil structural loads (BU-L1-A2). Bars: magnetic pressure $B^2/2\mu_0$ at the throat, plug and program fields against the 667 MPa allowable, with the reconciling three-dimensional FE peak (440 MPa, utilization 0.66) marked; line: the thick-wall Lamé required wall. The 26.49 T design point sits within allowable with margin.

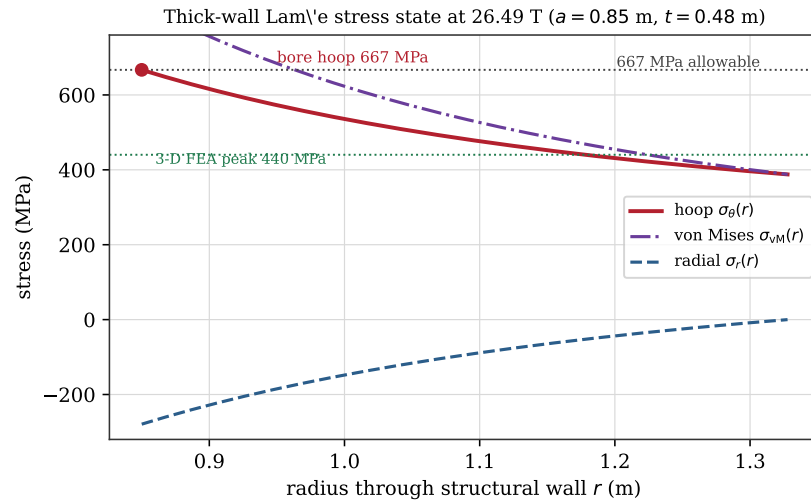


Figure 2. Thick-wall Lamé stress state through the structural overwrap at 26.49 T (equation 5; $a = 0.85$ m, $t = 0.48$ m). Hoop stress peaks at the bore at the 667 MPa allowable and falls outward; the von Mises equivalent tracks it; the radial stress runs from $-p_{\text{mag}}$ to zero. The reconciling three-dimensional FE peak (440 MPa) lies below the analytic bore value.

4.2 The conductor hoop stress, closed

The tightest structural quantity is not the coil shell but the conductor's own hoop stress (figure 3). The thin-wall screen of equation (7) finds the 17 T throat comfortable and the 26.49 T plug governing at ~ 814 MPa, a $0.86\times$ margin against the ~ 700 MPa reinforced allowable — feasible but reinforcement-limited, and flagged for FE closure (AG-CX-26). That closure, carried out on the real coil CAD with a full three-dimensional FE solver (P3-FEA-10), lifts the conductor margin to $1.34\times$, and the REBCO strain closes to a $2.85\text{--}3.54\times$ margin once a graded radial preload is applied — all with the plug dimensions unchanged. The thin-wall estimate was doing its job as a conservative screen; the real-CAD geometry and preload distribute the load and open the margin.

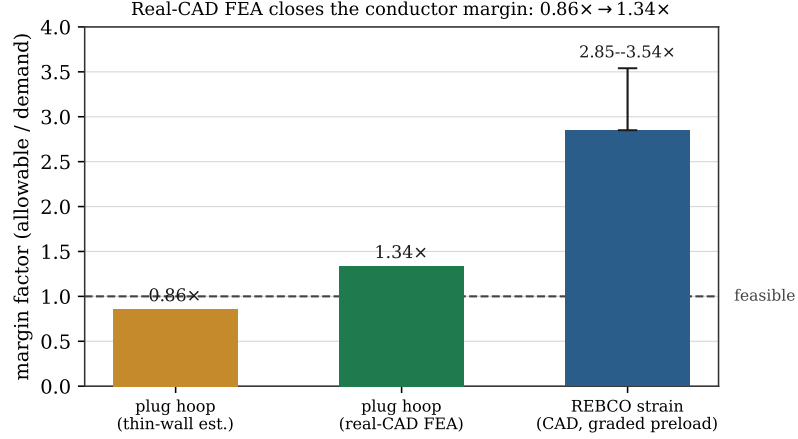


Figure 3. Conductor-margin closure. The thin-wall screen placed the 26.49 T plug hoop stress at a $0.86\times$ margin (AG-CX-26); the real-CAD three-dimensional FE analysis lifts it to $1.34\times$, and the REBCO strain reaches a $2.85\text{--}3.54\times$ margin under a graded preload (P3-FEA-10). Dashed line: feasibility ($1\times$).

4.3 Two magnets on separate load lines

Figure 4 places the plug and the centrepost on their own temperature-margin load lines (equation 8; KX-L1-A15). At its design current the plug holds $\Delta T \approx 37.1$ K at a load fraction $f_{\text{load}} \approx 0.43$ and 26.49 T; the separate centrepost holds $\Delta T \approx 49.5$ K at $f_{\text{load}} \approx 0.27$ and 16.84 T. Both magnets sit well short of their current-sharing lines, and each is modelled on its own critical surface — never the plug field substituted for the centrepost field or the reverse. The winding conductor basis behind the plug's ampere-turns is shown in figure 5 (SC-BU-1): an ampere-turn density of 13.5 MA m^{-1} and of order 4×10^2 km of REBCO tape per metre of plug length, large but not unprecedented for a high-field HTS solenoid.

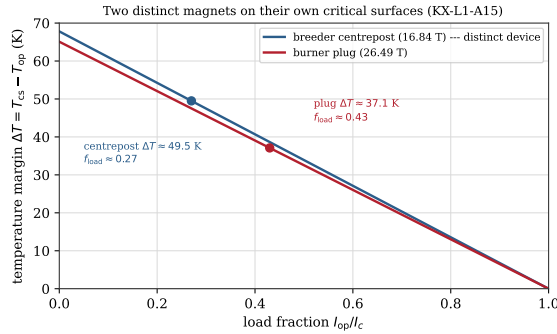


Figure 4. Temperature-margin load lines for the two distinct magnets (KX-L1-A15). Markers: the design current. Burner plug $\Delta T \approx 37.1$ K at 26.49 T; breeder centrepost $\Delta T \approx 49.5$ K at 16.84 T. Each is modelled on its own critical surface.

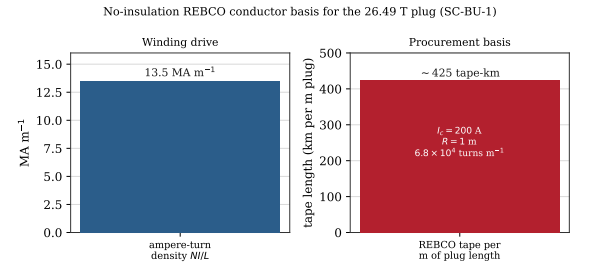


Figure 5. No-insulation REBCO conductor basis for the 26.49 T plug (SC-BU-1): ampere-turn density 13.5 MA m^{-1} and $\sim 4 \times 10^2$ km of tape per metre of plug length at $I_c = 200$ A, a 1 m winding radius and 6.8×10^4 turns m^{-1} .

4.4 Quench protection is sized inside the thermal limit

Figure 6 evaluates the adiabatic hot-spot integral of equation (10) for the copper-stabilized composite: the temperature reached rises with the load integral $Z = \int J_{\text{stab}}^2 dt$ and crosses 150 K at

$\Gamma(150\text{ K}) \approx 1.5 \times 10^{17} \text{ A}^2 \text{ s m}^{-4}$. The register's HX-10 detect-and-dump envelope — 4 s at 100 A mm^{-2} and 1.9 s at 200 A mm^{-2} , both keeping the hot spot below 150 K (figure 7) — corresponds to load integrals of 4.0 and $7.6 \times 10^{16} \text{ A}^2 \text{ s m}^{-4}$, comfortably inside $\Gamma(150\text{ K})$. The binding constraint is therefore the detection-and-dump architecture, not the material thermal limit; the plug has thermal headroom. Figure 8 shows the radial build — bore, NI-REBCO pack, structural overwrap — and figure 9 the detect-and-dump chain that fits inside the envelope; both are schematic. The fused, in-winding precursor that drives that chain is demonstrated separately [34], with a reduced-order twin reproducing the plug's operating anchors and quench margin (KX-L1-A15).

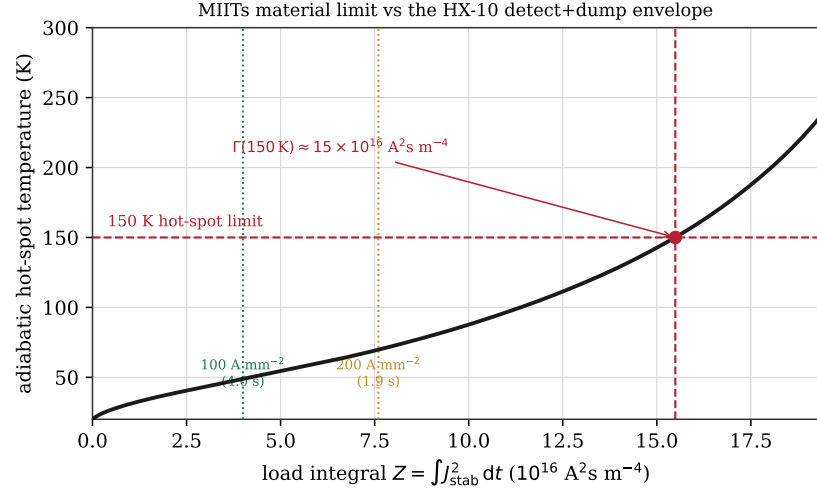


Figure 6. Adiabatic hot-spot (MIITs) material limit for the copper-stabilized composite ($\text{RRR} \approx 100$; equation 10). Temperature reached versus load integral $Z = \int J_{\text{stab}}^2 dt$; the 150 K limit corresponds to $\Gamma(150\text{ K}) \approx 1.5 \times 10^{17} \text{ A}^2 \text{ s m}^{-4}$. Vertical markers: the HX-10 detect-and-dump operating points, both inside the limit.

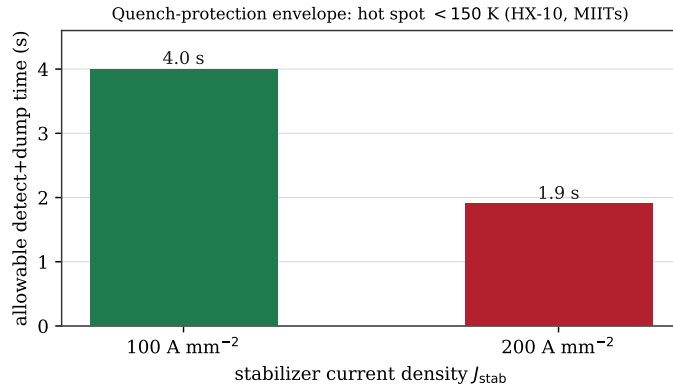


Figure 7. Quench-protection envelope (HX-10). Allowable detect-and-dump time to keep the adiabatic hot spot below 150 K, at two stabilizer current densities. The envelope is sized for the plug coil and lies inside the material limit of figure 6.

4.5 Qualification summary

Table 2 and figure 10 collect the qualification gates. The structural shell passes at a 0.66 utilization; the conductor hoop margin closes to $1.34\times$; the REBCO strain margin is $2.85\text{--}3.54\times$; the 17 T throat is comfortable; and the quench detect-and-dump envelope keeps the hot spot below 150 K within a multi-second dump. The single open item is a dedicated overfield structural-shell confirmation at $+20\%$ field (§6).

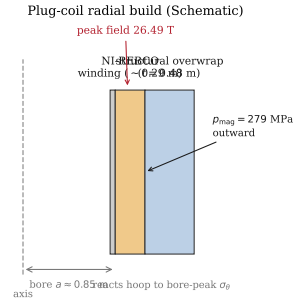


Figure 8. (Schematic.) Plug-coil radial build: bore ($a \approx 0.85$ m), NI-REBCO winding pack ($w \approx 0.29$ m) and structural overwrap ($t = 0.48$ m) reacting the 279 MPa magnetic pressure to a bore-peak hoop stress. No data values are plotted.

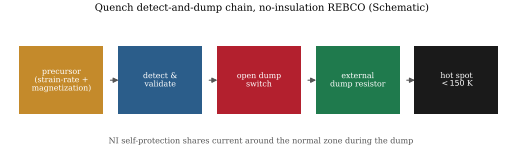


Figure 9. (Schematic.) Quench detect-and-dump chain for the no-insulation REBCO plug: non-voltage precursor, detect-and-validate, dump switch, external dump resistor, hot spot held below 150 K. NI self-protection shares current around the normal zone.

Table 2. Plug-coil (26.49 T) qualification gates and verdicts. Serials refer to the de-risking register [30].

gate	result	verdict
structural shell, three-dimensional FE (BU-L1-A2)	440 MPa peak, 0.66 utilization vs 667 MPa	PASS
conductor hoop, real-CAD FE (AG-CX-26→P3-FEA-10)	$0.86\times \rightarrow 1.34\times$ margin	PASS
REBCO strain, graded preload (P3-FEA-10)	$2.85\text{--}3.54\times$ margin	PASS
17 T mirror throat (AG-CX-26)	comfortable, non-binding	PASS
quench detect+dump, MIITs (HX-10)	1.9–4 s envelope, < 150 K	PASS
overfield shell +20% (BV-09)	confirming run set up	OPEN

Plug-coil qualification scorecard (26.49 T)

Structural shell (3-D FEA)	0.66 utilization vs 667 MPa	PASS
Conductor hoop (real-CAD FEA)	$1.34\times$ margin	PASS
REBCO strain (graded preload)	$2.85\text{--}3.54\times$ margin	PASS
17 T mirror throat	comfortable	PASS
Quench detect+dump (< 150 K)	1.9–4 s envelope	PASS
Overfield shell (+20%)	confirming run set up	OPEN

Figure 10. Plug-coil qualification scorecard at 26.49 T. Every structural and protection gate passes with margin; the single open item is the overfield shell confirmation (BV-09).

5 Discussion

5.1 Against the high-field-magnet literature

The plug coil is a 26.49 T REBCO magnet, and its structural numbers should be read against the fields the community has demonstrated. The 45.5 T REBCO-insert record [1] and the 32 T all-superconducting user magnet [2] both operate above the plug’s peak field, so the conductor and the field regime are not the frontier; the engineering question is the mechanical and protection design at fusion scale, which the SPARC toroidal-field model-coil program addressed for a REBCO magnet at reactor relevance [9, 8]. The plug’s 279 MPa magnetic pressure and 667 MPa structural allowable are conventional for high-field superconducting magnet structures [11, 12], and the reconciling FE utilization of 0.66 leaves the $1.5\times$ margin that magnet practice expects. The conductor hoop-stress closure is the substantive result: coated-conductor tapes tolerate large axial stress but are strain- and delamination-sensitive [17, 4], so the move from the $0.86\times$ thin-wall screen to the $1.34\times$ real-CAD margin — with a $2.85\text{--}3.54\times$ strain margin under graded preload — is exactly the kind of load-sharing and preload engineering that qualifies a REBCO insert. The message is not that the plug reaches a record field; it is that at a field the community has already exceeded, the Kronos plug is engineered to margin in structure, conductor stress and strain simultaneously.

5.2 Against the quench-protection literature

The protection case rests on the slow-NZPV physics of no-insulation REBCO. Marchevsky’s review makes the core point quantitatively: in HTS conductors the large temperature margin and slow propagation mean a hot spot can reach damaging temperature before a measurable resistive voltage appears, so voltage detection alone is inadequate [14]. The community response has been non-voltage observables — fibre-Bragg thermometry demonstrated on VIPER cable at SULTAN [15] and current-distribution monitoring on fusion cables [16]. The plug’s MITs envelope (equation 11) is consistent with this picture: the $1.5 \times 10^{17} \text{ A}^2 \text{ s m}^{-4}$ material limit gives multi-second detect-and-dump headroom at $100\text{--}200 \text{ A mm}^{-2}$, comfortably longer than the microsecond-to-second response times those non-voltage detectors report, so the binding constraint is architectural rather than thermal. Matching the plug to the fused in-winding precursor demonstrated for the Kronos NI-REBCO magnets [34] closes the loop: a detector whose lead time is set by the millisecond NZPV crossing fits inside a multi-second dump envelope with room to spare.

5.3 Why the separation of the two magnets matters

Keeping the plug (26.49 T) and centrepole (16.84 T) on separate load lines (figure 4) is not bookkeeping. The two magnets differ in peak field by more than 9 T, in load fraction (0.43 vs 0.27), and in temperature margin (37.1 vs 49.5 K), and they belong to two different products — the tandem-mirror burner and the spherical-tokamak breeder [32, 31]. Conflating their fields would either over-design the centrepole or, more dangerously, under-design the plug. The magnet-integrity case requires that each be qualified on its own critical surface, which is what this paper and its companions do.

6 Limitations

One honest gate remains. The reconciling and real-CAD FE analyses qualify the coil at the 26.49 T design point, but a dedicated three-dimensional structural-shell check at $+20\%$ overfield — the standard proof-margin bound — is the one confirming analysis still to run. It is set up as an **sfepy** three-dimensional model (BV-09) and is the overfield bound on the qualified design; the structural headroom already demonstrated to 30 T (figure 1) makes it a confirmation rather than an open risk, but it is not yet a frozen result and is reported as such.

7 Conclusion

The Kronos burner plug coil is a qualified 26.49 T no-insulation REBCO magnet. Its structure carries the 279 MPa Maxwell magnetic pressure at a 0.66 utilization of the 667 MPa allowable, with a thick-wall Lamé overwrap of 0.48 m at a 0.85 m bore and demonstrated headroom to 30 T; the conductor hoop-stress margin, $0.86\times$ on a conservative thin-wall screen, closes to $1.34\times$ on the real CAD, with a $2.85\text{--}3.54\times$ REBCO strain margin under a graded preload and unchanged plug dimensions; and an adiabatic hot-spot analysis places a 1.9–4 s detect-and-dump envelope, keeping the hot spot below 150 K, comfortably inside a $1.5 \times 10^{17} \text{ A}^2 \text{ s m}^{-4}$ material limit. The 17 T mirror throat is never binding, and the burner plug (26.49 T), kept distinct from the breeder centrepole (16.84 T) throughout, is a structurally sound, protectable, buildable high-field magnet. Because the

magnet that sets the plug density ratio is the pacing subsystem of the burner, its qualification is a load-bearing step toward the D-³He tandem-mirror product [37, 36, 35].

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Reproducibility. The structural loads, the thick-wall Lamé sizing, the reconciling and real-CAD finite-element analyses, and the quench envelope are frozen anchors (BU-L1-A2, AG-CX-26, P3-FEA-10, HX-10, HX-11, SC-BU-1, KX-L1-A15) in the Kronos Physics De-Risking Register and are regenerable from the analytic, CalculiX and MIITs scripts archived with this paper. This paper is archived at doi:10.5281/zenodo.22645915; official version:

<https://www.kronosfusionenergy.com/publications/engineering-the-26-49-t-plug-coil-of-a-d-he-tandem-mirr.html>, and the register at doi:10.5281/zenodo.22133057.

Data availability

This paper is archived at doi:10.5281/zenodo.22645915. The structural tables, finite-element outputs, conductor critical-surface load lines, and quench envelope are archived with the deposit and reference the Kronos 2026 series. The figure-generation script that produced every data figure is deposited with the manuscript. Any economic analysis is reported separately and is not included in the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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